

Callum J. Hodge

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EDUCATION:

University of Warwick Coventry, UK

(Sep. 2023 - June 2027)

MPhys Physics with Astrophysics

Current Weighted Average mark: 91% (1st)

Top Module Marks

Electrodynamics:	99%	Computational Physics:	98%
Black Holes, White Dwarfs & Neutron Stars:	96%	Mathematics for Physicists:	96%
Quantum Physics of Atoms:	96%	Astrophysics Laboratory II and Skills:	85%

PUBLICATIONS & RESEARCH EXPERIENCE:

First Author on Research Paper submitted to *Monthly Notices of the Royal Astronomical Society* (June – Aug. 2026)

- 8-week research project determining the first estimate for the spin-down rate of a special white dwarf, in one of only three known white dwarf pulsars (a type of binary system that undergoes periodic, pulsed emission).
- Funded by winning a Royal Astronomical Society undergraduate research grant.
- Full research paper: (currently submitted and under peer-review).

Summer Research with the *University of Warwick Undergraduate Research Support Scheme* (June – Aug. 2025)

- 8-week research project studying a phenomenon known as the *Brown Dwarf Desert*, where brown dwarfs orbiting main sequence stars are largely observed at wide separations. Project involved python simulations.
- Funded by winning an undergraduate research grant from the University of Warwick.
- Full research paper: <https://urss.warwick.ac.uk/items/show/806>.

Conference Presenting at the *British and International Conferences of Undergraduate Research* (April & Sep. 2026)

- Presenting my own astrophysics research in poster and presentation formats to other undergraduate students from across the country for a total of four days across both conferences.
- Astrophysics research presented was from both projects listed above.

Final-year Undergraduate Research Project (Ongoing) (Sep. 2026 – Present)

- Using radio observations to track the blast-wave of a supernova. This gives us insight into the properties of the surrounding medium and allows us to track the history of mass-loss of the stellar progenitor.

AWARDS & HONOURS:

Physics Distinction List for Third Year

(Sep. 2025 – July 2026)

- Awarded to the top 10% of each cohort every year. Recognition of academic excellence in physics.

Duke of Edinburgh Award (Bronze and Gold)

(Sep. 2019 – Aug. 2022)

- National youth development award that builds skills via volunteering, physical activity and skills development.

Warwick Award: Gold

(Sep. 2023 – Sep. 2024)

- Awarded for proficiency in core employability skills from 300 hours of academic and extra-curricular activities.

- Skills include critical thinking, organizational awareness, problem solving, teamwork and professionalism.

COMPUTING & LABORATORY SKILLS:

- High proficiency in Python, C and LaTeX. Average mark of 94% across three years of Python/C coding modules.
- Analysed high-time resolution photometric data from ULTRACAM for my project on white dwarf pulsars (above).
- Extensive, hands-on experience using equipment such as telescopes, spectrographs and electronics from lab modules. Average mark of 79% across three years of lab modules.
- Experience using astrophysics software such as SAO ds9, Aperture Photometry Tool, VIREO and FireCapture.

SERVICE & OUTREACH:

Revision Lectures

(May 2024 – May 2026)

- Delivery of nine hours' worth of revision lectures to over 100 other undergraduate students.
- Content consisted of that from the modules *Classical Mechanics & Special Relativity*, *Quantum Mechanics & Its Applications*, *Stars & The Solar System* and *Black Holes, White Dwarfs & Neutron Stars*.

Volunteering with the University of Warwick Mobile Planetarium

(June 2026 – Present)

- Delivering educational shows to schools and youth organisations on astronomy, space exploration and the importance of science.

Live Labs and Resonate Roadshow

(Dec. 2023 & Dec. 2025)

- Volunteering at events held to encourage local people and school children to immerse themselves in science.
- Running workshops that create visual experiences for the general public on different topics in astrophysics.

University of Warwick Department of Physics Student Ambassador

(Sep. 2021 – Aug. 2023)

- Working on University open days to give tours of the campus, deliver information talks and answer questions.
- Communicating and presenting for tours and talks, managing groups of prospective students, enacting professionalism and key decision-making when answering questions and giving advice.

Maths Mentoring

(Jan. – May 2024)

- Volunteering as a mentor for two students taking GCSE maths.
- Organising and directing weekly revision sessions, as well as writing and marking weekly homework sheets.

EXTRA-CURRICULAR ACTIVITIES:

Warwick University Formula 1 Society Executive

(April 2024 – April 2026)

President, Treasurer and Content Creation Officer

- Role involves organising society events, leading society executive team, managing finances and social platforms.
- Delegating tasks to other executive members, balancing this workload with my degree, communicating and collaborating with external agencies, being a team player in my society executive team.

TeamWork 2024 International Experience Programme

(June – July 2024)

- Participation in 4-week research project with overseas students, working for McDermott International. Project aimed to evaluate different types of renewable energy sources to advise the company on future investment.
- Taking leadership over a project, maintaining punctuality to work towards deadlines, collaborating and communicating with members of my team effectively, dealing with complex issues in our research.